

Quantium Virtual Internship

Retail Strategy and Analytics — Task 1

Chip Category Review: Data Preparation & Customer Analytics

Prepared for: Julia (Category Manager) & Zilinka

Datasets: QVI_transaction_data.xlsx, QVI_purchase_behaviour.csv

Analysis conducted in Python (pandas, matplotlib, scipy)

1. Introduction & Objective

Julia, the Category Manager for chips, needs an initial, data-backed view of who buys chips and what drives their purchasing, ahead of the upcoming category review. This document walks through the data checks, cleaning, feature engineering and customer segment analysis performed on the transaction and customer datasets, and closes with commercial recommendations.

2. Loading the Data

```
import pandas as pd

transactionData = pd.read_excel('QVI_transaction_data.xlsx')
customerData = pd.read_csv('QVI_purchase_behaviour.csv')

print(transactionData.shape)  # (264836, 8)
print(customerData.shape)    # (72637, 3)
```

The transaction data contains 264,836 rows across 8 columns (date, store, loyalty card, transaction ID, product number/name, quantity and total sales). The customer data contains 72,637 rows mapping each loyalty card number to a LIFESTAGE (e.g. young families, retirees) and a PREMIUM_CUSTOMER tier (Budget / Mainstream / Premium).

3. Exploratory Data Analysis & Cleaning

3.1 Date format

```
# DATE is stored as an integer day-count (Excel/CSV serial date).
# Excel's epoch begins 30 Dec 1899.
transactionData['DATE'] = pd.to_datetime(transactionData['DATE'], unit='D', origin='1899-12-30')
print(transactionData['DATE'].min(), transactionData['DATE'].max())
# 2018-07-01 to 2019-06-30
```

The data spans exactly one year: 1 July 2018 to 30 June 2019.

3.2 Checking PROD_NAME — are these all chips?

```
import re
from collections import Counter

all_words = []
for name in transactionData['PROD_NAME'].unique():
    all_words.extend(name.split())

words_no_digits = [w for w in all_words if not any(ch.isdigit() for ch in w)]
words_clean = [re.sub(r'^A-Za-z|', '', w) for w in words_no_digits]
words_clean = [w for w in words_clean if w != '']
Counter(words_clean).most_common(10)
```

Top words: Chips (21), Smiths (16), Crinkle (14), Cut (14), Kettle (13), Cheese (12), Salt (12), Original (10), Chip (9), **Salsa (9)**. The word "Salsa" confirms there are 9 distinct Salsa dip products mixed into the dataset (18,094 transaction rows) — these are not chips, so they are removed.

```
transactionData['SALSA'] = transactionData['PROD_NAME'].str.lower().str.contains('salsa')
transactionData = transactionData[transactionData['SALSA'] == False].drop(columns=['SALSA'])
# Shape after removing salsa: (246742, 8)
```

3.3 Outlier check — 200 packets in one transaction

```
transactionData['PROD_QTY'].describe()
# mean 1.9, 75th pct 2, max 200 -> clear outlier
```

```

outlier = transactionData[transactionData['PROD_QTY'] >= 200]
transactionData[transactionData['LYLTY_CARD_NBR'] == outlier['LYLTY_CARD_NBR'].iloc[0]]

```

Two transactions of exactly 200 packets (\$650 each) were both made by loyalty card **226000**, and these are the *only* two transactions that customer made all year. This is not an ordinary household shopping pattern — it looks like a commercial/bulk buyer rather than a retail customer, so this loyalty card is removed from further analysis.

```

transactionData = transactionData[transactionData['LYLTY_CARD_NBR'] != 226000]
# Shape after removing customer 226000: (246740, 8); max PROD_QTY is now 5

```

3.4 Checking for missing dates

```

txn_by_date = transactionData.groupby('DATE').size().reset_index(name='N')
print(txn_by_date.shape[0]) # 364 -> one day is missing from the expected 365

```

```

full_dates = pd.DataFrame({'DATE': pd.date_range('2018-07-01', '2019-06-30')})
transactions_by_day = full_dates.merge(txn_by_date, on='DATE', how='left').fillna(0)
transactions_by_day[transactions_by_day['N'] == 0] # 2018-12-25

```

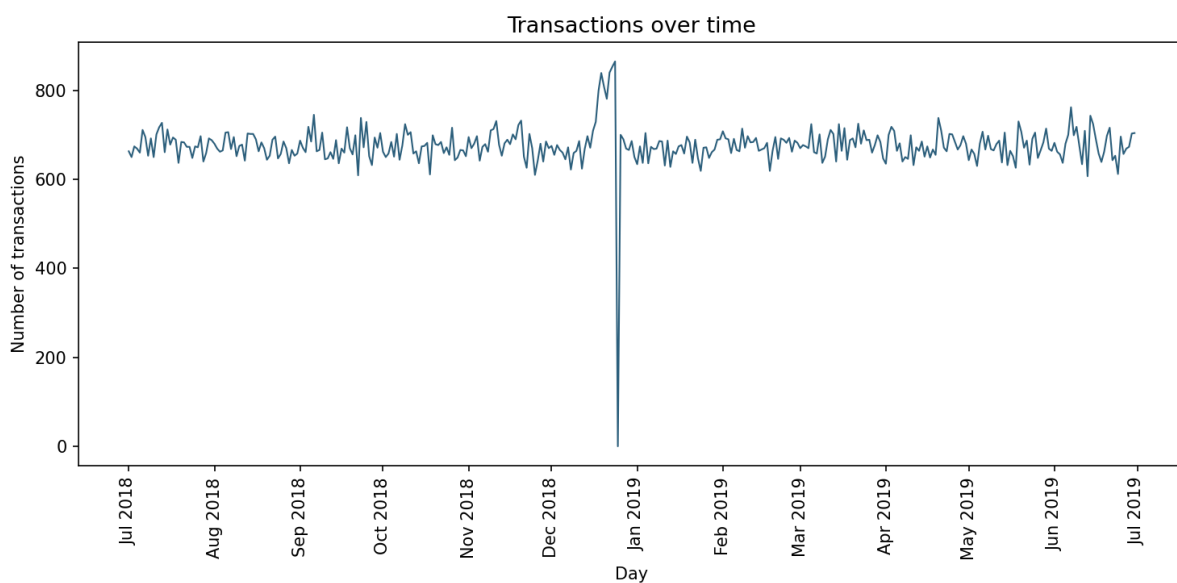


Figure 1. Daily number of transactions, 1 Jul 2018 – 30 Jun 2019.

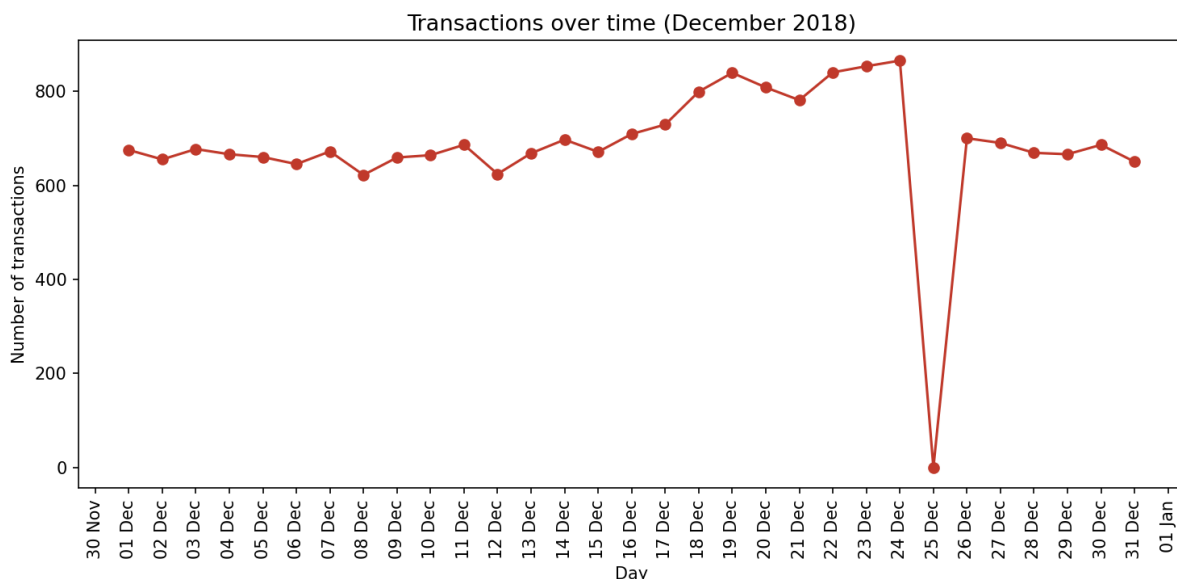


Figure 2. December zoom — sales build steadily toward Christmas, then drop to zero on 25 Dec.

Only 364 of the expected 365 days appear in the data — **25 December 2018 (Christmas Day)** has zero transactions, consistent with stores being closed. Sales ramp up steadily in the two weeks before Christmas and then resume at normal levels on Boxing Day, confirming a genuine seasonal pattern rather than a data quality fault.

4. Feature Engineering: Pack Size & Brand

4.1 Pack size

```
transactionData['PACK_SIZE'] = transactionData['PROD_NAME'].str.extract(r'(\d+)').astype(int)
transactionData.groupby('PACK_SIZE').size().sort_index()
# smallest 70g, largest 380g -- sensible range for chip packs
```

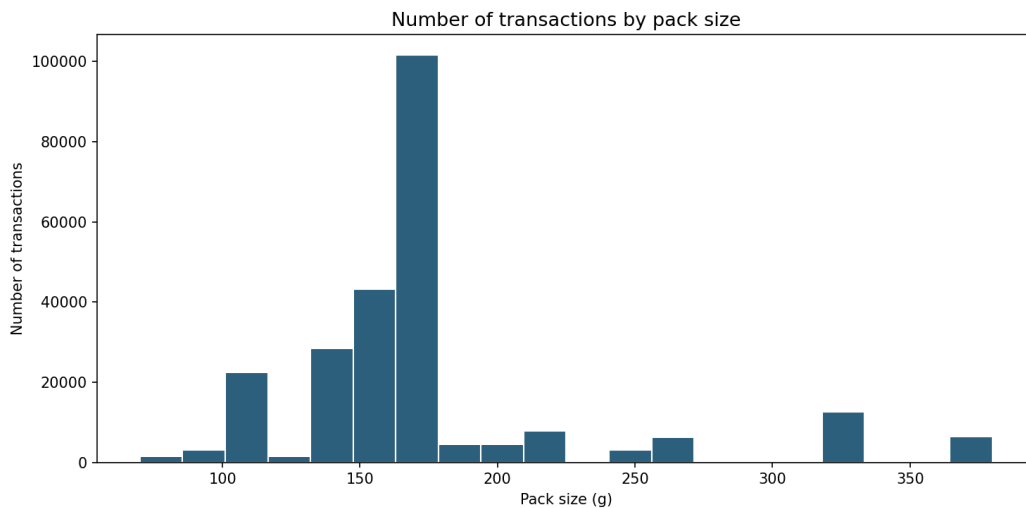


Figure 3. Distribution of transactions by pack size (g). 175g is the most common size.

4.2 Brand

```
transactionData['BRAND'] = transactionData['PROD_NAME'].str.split().str[0].str.upper()

# Consolidate near-duplicate / abbreviated brand names referring to the same brand
brand_map = {
    'RED': 'RRD', 'SNBTS': 'SUNBITES', 'INFZNS': 'INFUZIONI', 'WW': 'WOOLWORTHS',
    'SMITH': 'SMITHS', 'NCC': 'NATURAL', 'DORITO': 'DORITOS',
    'GRAIN': 'GRAINWAVES', 'GRNWVES': 'GRAINWAVES',
}
transactionData['BRAND'] = transactionData['BRAND'].replace(brand_map)
```

For example "RED" and "RRD" both refer to Red Rock Deli, and "GRAIN"/"GRNWVES" both refer to Grain Waves — these are consolidated so each brand is counted once. After cleaning there are 20 distinct brands, led by Kettle (41,288 units sold), Smiths (30,353) and Doritos (25,224).

5. Merging Transaction & Customer Data

```
data = transactionData.merge(customerData, on='LYLTY_CARD_NBR', how='left')
print(transactionData.shape[0], data.shape[0]) # 246740, 246740 -- no duplicates created
data.isnull().sum() # 0 nulls across all columns
```

The merged dataset has the same number of rows as the cleaned transaction data, confirming a clean one-to-one join with no duplication, and there are zero unmatched (null) records — every transaction's loyalty

card was found in the customer dataset.

6. Customer Segment Analysis

With the data cleaned and enriched, we can now look at who is driving chip sales, broken down by LIFESTAGE and PREMIUM_CUSTOMER tier.

6.1 Total sales by segment

```
sales_seg = data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['TOT_SALES'].sum()
pivot_sales = sales_seg.unstack('PREMIUM_CUSTOMER')[['Budget', 'Mainstream', 'Premium']]
```

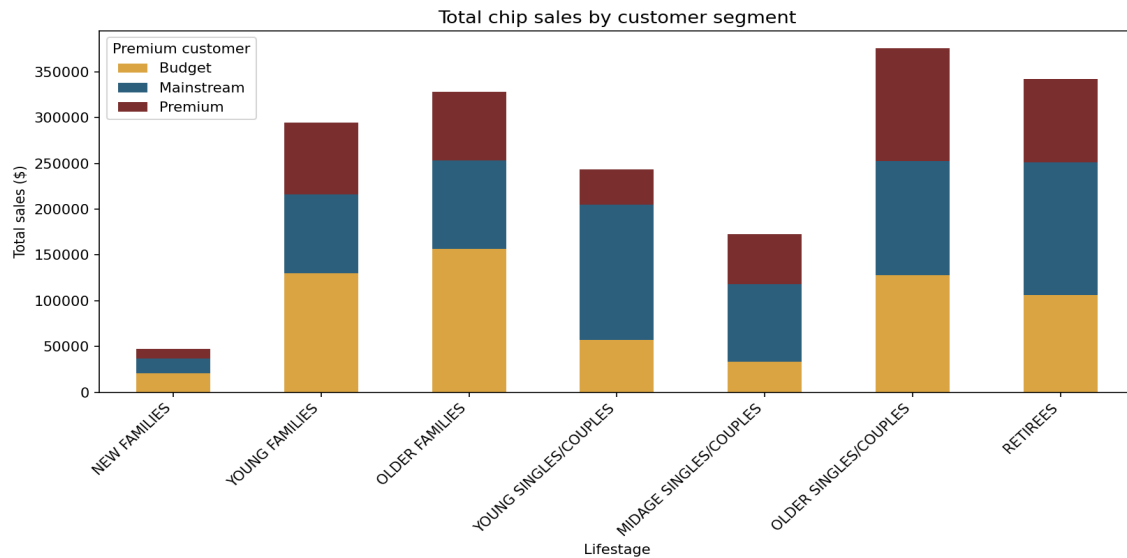


Figure 4. Total chip sales (\$) by lifestage and premium tier.

The three largest contributors to total chip sales are **Mainstream — young singles/couples** (~\$147.6k), **Budget — older families** (~\$156.9k) and **Mainstream — retirees** (~\$145.2k).

6.2 Number of customers by segment

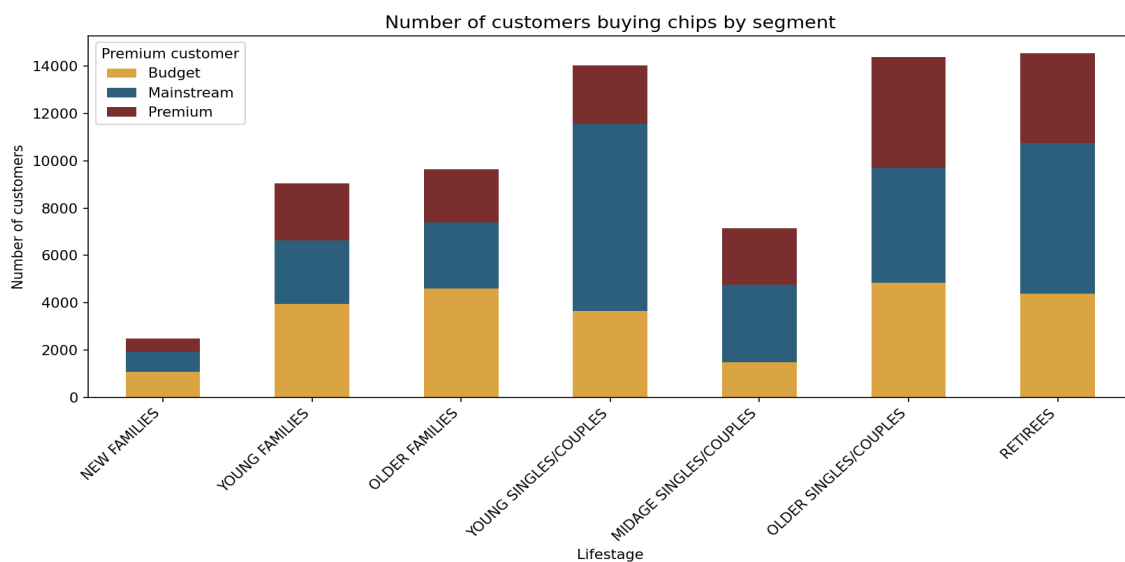


Figure 5. Number of distinct customers buying chips, by lifestage and premium tier.

Mainstream young singles/couples (7,917 customers) and Mainstream retirees (6,358) are also the largest customer *populations* — so their high sales are partly explained by there simply being more of them. Budget older families, by contrast, drive high sales with a comparatively smaller customer base (4,611 customers), implying each of these households buys more chips on average.

6.3 Average units per customer

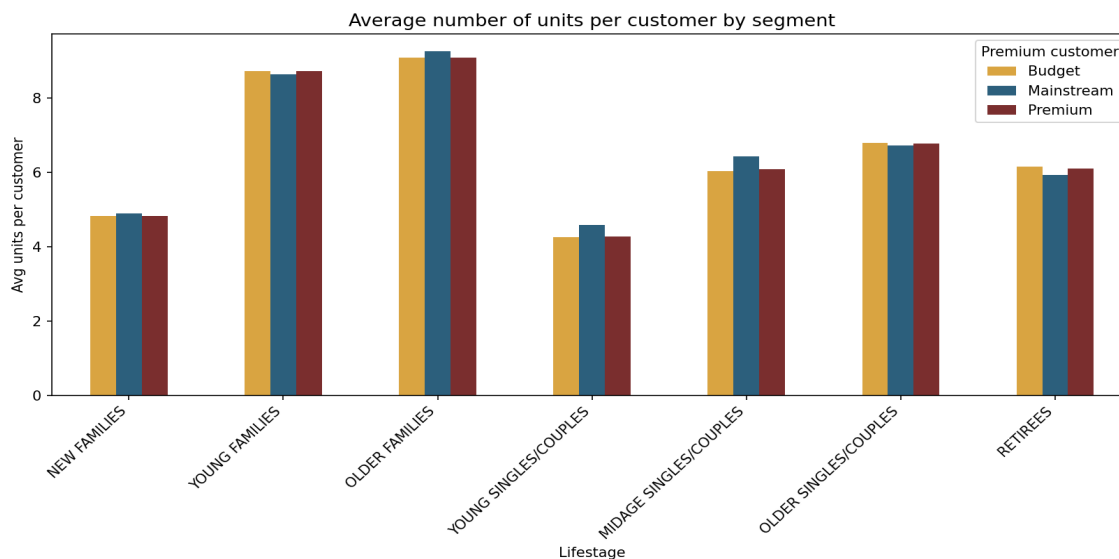


Figure 6. Average number of chip units bought per customer, by segment.

Older and young families buy the most units per customer (8.6–9.3 units), regardless of premium tier — consistent with feeding a household rather than individual snacking. Young singles/couples buy the fewest units per customer (4.3–4.6), which fits a smaller-household, more occasional purchase pattern.

6.4 Average price per unit

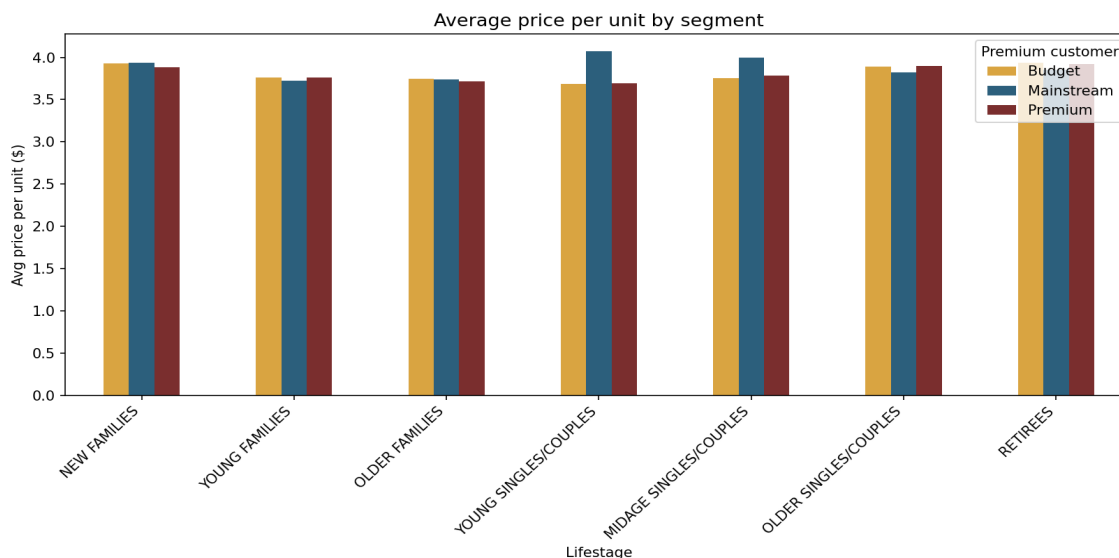


Figure 7. Average price paid per unit (\$), by segment.

Mainstream young singles/couples and Mainstream midage singles/couples pay noticeably more per unit (~\$4.07 and ~\$3.99) than their Budget or Premium counterparts in the same life stage (~\$3.69–\$3.78). This is a distinctive

pattern: in most other lifestages, Premium customers pay a similar or higher price than Mainstream, but among singles/couples the Mainstream segment is the one paying the premium price.

6.5 Statistical test: is the price difference significant?

```
from scipy import stats

target_lifestages = ['YOUNG SINGLES/COUPLES', 'MIDAGE SINGLES/COUPLES']
subset = data[data['LIFESTAGE'].isin(target_lifestages)]
data['PRICE_PER_UNIT'] = data['TOT_SALES'] / data['PROD_QTY']

mainstream_price = subset.loc[subset['PREMIUM_CUSTOMER']=='Mainstream', 'PRICE_PER_UNIT']
other_price = subset.loc[subset['PREMIUM_CUSTOMER'].isin(['Budget', 'Premium']), 'PRICE_PER_UNIT']

t_stat, p_val = stats.ttest_ind(mainstream_price, other_price, equal_var=False)
```

Result: Mainstream mean price/unit = \$4.040 (n=30,639) vs Budget/Premium mean price/unit = \$3.706 (n=26,728). $t = 37.62$, $p < 0.0001$.

The p-value is far below 0.05, so the unit price for Mainstream young and midage singles/couples **is** significantly higher than that of their Budget or Premium counterparts. This is a robust, not coincidental, willingness to pay more — a strong commercial signal.

7. Deep Dive: Mainstream Young Singles/Couples

This segment combines high customer numbers, high total sales and a willingness to pay more per unit — making it a priority target. We dig into brand and pack-size preference using an affinity index (the segment's share of purchases for an item, divided by the same share in the rest of the population; above 1.0 means the segment over-indexes on that item).

7.1 Brand affinity

```
target = data[(data['LIFESTAGE']=='YOUNG SINGLES/COUPLES') & (data['PREMIUM_CUSTOMER']=='Mainstream')]
rest   = data[~((data['LIFESTAGE']=='YOUNG SINGLES/COUPLES') & (data['PREMIUM_CUSTOMER']=='Mainstream'))]

target_share = target.groupby('BRAND')['PROD_QTY'].sum() / target['PROD_QTY'].sum()
rest_share   = rest.groupby('BRAND')['PROD_QTY'].sum() / rest['PROD_QTY'].sum()
brand_affinity = (target_share / rest_share).sort_values(ascending=False)
```

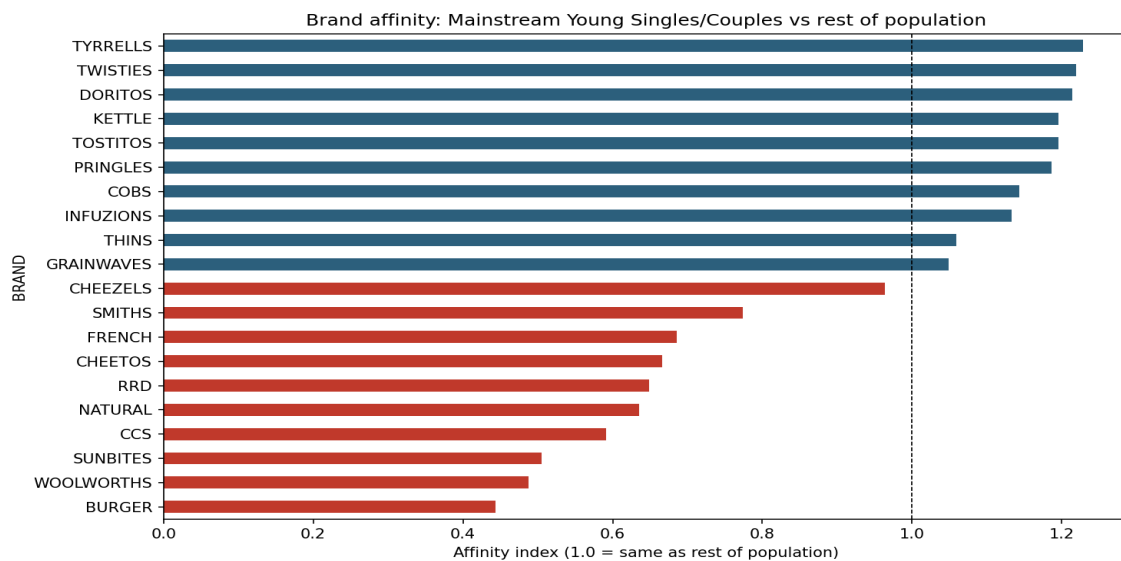


Figure 8. Brand affinity index, Mainstream young singles/couples vs rest of population.

We can see that: Mainstream young singles/couples over-index most strongly on **Tyrrells (1.23x)**, **Twisties (1.22x)**, **Doritos (1.22x)**, **Kettle (1.20x)** and **Tostitos (1.20x)** — leaning towards premium/artisan and bold-flavoured brands. They under-index heavily on **Burger Rings (0.44x)**, **Woolworths own-brand (0.49x)** and **Sunbites (0.51x)**, suggesting limited interest in value own-label or better-for-you snack options.

7.2 Pack size affinity

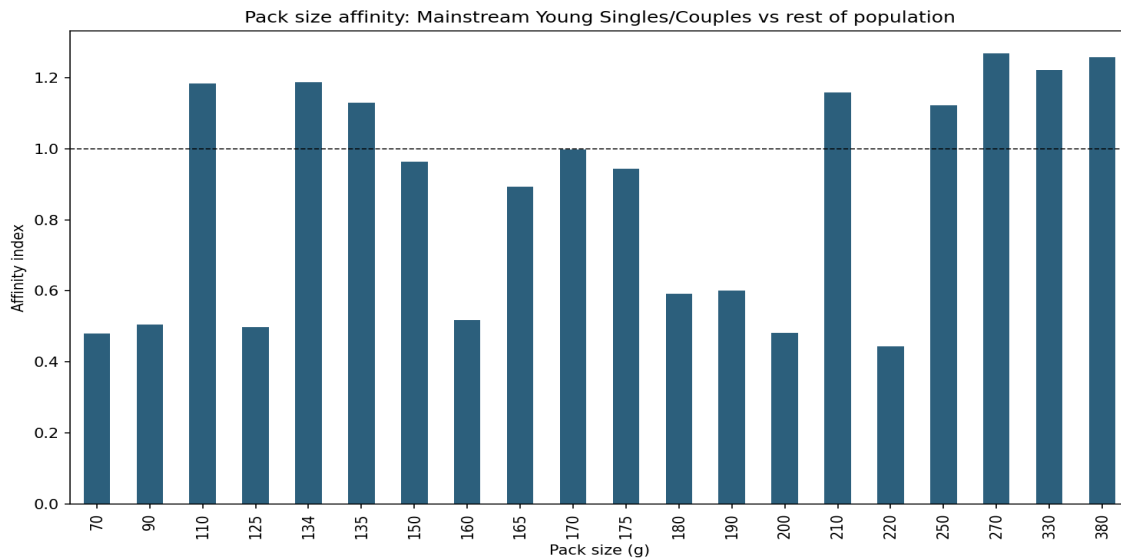


Figure 9. Pack size affinity index, Mainstream young singles/couples vs rest of population.

We can see that: this segment over-indexes most on large/sharing packs — **270g (1.27x)**, **380g (1.26x)** and **330g (1.22x)** — and under-indexes on small packs (70g–220g, affinity 0.44x–0.60x). The only brand selling a 270g pack is **Twisties**, which combines both the top brand and top pack-size preferences for this segment into a single, highly targeted product.

8. Strategic Recommendations for Julia

These findings point to a clear, commercially actionable strategy for the category review:

1. Prioritise Mainstream — Young Singles/Couples and Mainstream — Retirees

These are the two largest customer bases buying chips and the two largest sales contributors. Together they represent the biggest opportunity for incremental volume from acquisition or frequency growth.

2. Lean into bigger, bolder, sharing-format packs for the Mainstream young singles/couples segment

This segment already pays more per unit and over-indexes on large packs (270g–380g) and bold/premium brands (Tyrrells, Twisties, Doritos, Kettle, Tostitos). Featuring and promoting large-format SKUs of these brands — e.g. ranging Twisties 270g prominently, and considering similar large-pack variants from Tyrrells/Doritos/Kettle — should convert existing willingness-to-pay into higher basket value, with low risk of price pushback given the segment already pays a premium price point.

3. Treat Budget — Older Families and Older Families generally as a volume, not premium, opportunity

These households buy the most units per customer but at lower per-unit prices. Multi-buy or larger value-pack promotions (rather than premium-brand pushes) are likely to resonate better and drive volume without competing on margin in a segment that is price sensitive.

4. Re-test pricing/promotion for Mainstream midage and young singles/couples

The statistically significant price premium this group already pays ($p < 0.0001$) suggests room to trial modest price or mix optimisation (e.g. shifting promotional depth from value packs to premium/large packs) without harming volume.

5. Operational note

Exclude loyalty card 226000 from any customer-level modelling — it represents bulk/commercial buying behaviour, not a representative retail household, and would distort any per-customer averages if left in.

Suggested next step: validate the large-pack/premium-brand hypothesis for Mainstream young singles/couples with a small in-store or promotional trial (e.g. a featured end-cap of Twisties 270g and a large-format Tyrrells/Doritos SKU) in a sample of stores with a high concentration of this segment, and measure uplift in sales and units per customer before rolling out more broadly.